

Lab for randomized block trials assigning clusters within blocks

Chapter 10

Case Study

High School After-School Robotics Program

Dr. Carlos – a computer scientist – and his students have been running an after-school outreach program focused on robotics at a nearby high school in Atlanta. This high school predominately serves Black students – which are underrepresented in computer science – and the students come from both middle-class and low-SES families. In order to be admitted to the program, which is free, students had to apply and be recommended by a teacher. Once admitted, students in the program meet once a week for 2 hours for 16 weeks. During the lessons, they learn to program and build robots to complete tasks and, at the end of the program, they take part in a mini ‘competition’. The research team has been excited to see how the students have responded to the program, including how this has affected their interest in computer science in general, as well as in studying a STEM related field in college. Teachers have reported that students in the program are more engaged in their science classes.

Plausible Evaluation Design

Random Assignment of Clusters Within Blocks (Schools): Fixed School Effects

Suppose researchers wish to evaluate the effectiveness of a similar program on student science grades in school. Researchers enroll teachers and their students in the study during the the fall term. Teachers who are enrolled into the study are randomized within each school into either the treatment condition (the program) or a control condition. Spring science scores on a state-mandated standardized test are used as the outcome. A set of schools are recruited for the study from a variety of districts through a professional development conference for state principals and district membership is not considered in the design. For planning purposes, researchers use a teacher-level (within-school) intraclass correlation of .05.

The analysis will employ a two-level hierarchical linear model (students nested within teachers) with a random intercept at the teacher level and block covariates for schools to compare the averages of the treatment and control students.

For simplicity, assume perfect compliance with random assignment and a perfect response rate.

Question 1

Using the planner tool, find the minimum detectable effect for a randomized block study that employed 5 schools, in each of which are 2 teachers randomized into treatment and 2 randomized into control, each with 20 students, when power is .8 for a test with a significance level of .05.

```
compute_mdes(V = blockedclustered_fixed_nocovariate_V(m = 5,
                                                         kt = 2,
                                                         kc = 2,
                                                         n = 20,
                                                         icc = .05),
              df = blockedclustered_fixed_nocovariate_df(m = 5,
                                                         kt = 2,
```

```

                                kc = 2,
                                n = 20),
    beta = 1-.8,
    alpha = .05
)

## [1] 0.5305286

```

Question 2

A member of the study team claims that many schools only have a two teachers. Using the planner tool, find the minimum detectable effect for a randomized block study that employed 5 schools, in each of which there is 1 teacher randomized into treatment and 1 randomized into control, each with 20 students, when power is .8 for a test with a significance level of .05.

```

compute_mdes(V = blockedclustered_fixed_nocovariate_V(m = 5,
                                                        kt = 1,
                                                        kc = 1,
                                                        n = 20,
                                                        icc = .05),
             df = blockedclustered_fixed_nocovariate_df(m = 5,
                                                         kt = 1,
                                                         kc = 1,
                                                         n = 20),
             beta = 1-.8,
             alpha = .05
)

## [1] 0.7502808

```

Question 3

The results from Question 2 are different from Question 1. If the students and teachers are fixed, schools may need to be added. How many schools need to be added to the design in Question 2 to get a similar MDES as what you found for Question 1? Before you answer, consider whether the total sample sizes in Questions 1 and 2, would you need to increase blocks proportionally?

```

compute_mdes(V = blockedclustered_fixed_nocovariate_V(m = 8,
                                                        kt = 1,
                                                        kc = 1,
                                                        n = 20,
                                                        icc = .05),
             df = blockedclustered_fixed_nocovariate_df(m = 8,
                                                         kt = 1,
                                                         kc = 1,
                                                         n = 20),
             beta = 1-.8,
             alpha = .05
)

## [1] 0.5118141
print(paste("with 8 schools"))

## [1] "with 8 schools"

```

Question 4

After signing up the schools, it was apparent that using a 10th grade math test as a covariate was possible for the analysis. Earlier analyses of other cohorts show that the 10th grade math test was correlated with the focal science test at the teacher level at $\rho = .8$ for an R-square value of $\rho^2 = .64$, and at the student level at $\rho = .5$ for an R-square value of $\rho^2 = .25$. With this covariate, would it be possible to still use the sample design in Question 2 to get the sensitivity of plan in Question 1?

Using the planner tool, find the minimum detectable effect for a randomized block study with random effects that employed 5 schools, in each of which there is 1 teacher randomized into treatment and 1 randomized into control, each with 20 students, and using the covariate, when power is .8 for a test with a significance level of .05.

```
compute_mdes(V = blockedclustered_fixed_covariate_V(m = 5,
                                                    kt = 1,
                                                    kc = 1,
                                                    n = 20,
                                                    icc = .05,
                                                    R2_1 = .25,
                                                    R2_2 = .64
                                                    ),
            df = blockedclustered_fixed_covariate_df(m = 5,
                                                    kt = 1,
                                                    kc = 1,
                                                    n = 20,
                                                    q = 1),
            beta = 1-.8,
            alpha = .05
        )
```

```
## [1] 0.6346006
```

Is this covariate enough? What other adjustments could be made?

References

Barker, B. S., & Ansorge, J. (2007). Robotics as means to increase achievement scores in an informal learning environment. *Journal of research on technology in education*, 39(3), 229-243.